

Differentiation and Integration over Experiential Time

A mathematical companion to *The Resonating Self* — from the difference quotient to the PPTQ-scaled Kuramoto model

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Contents

1	Preface: why this manual exists	4
1.1	The three books of Steven Strogatz as reference and source of inspiration	4
1.2	How to read this manual	4
2	PART I — THE GUIDING THREAD: WITH RESPECT TO WHAT DO WE DIFFERENTIATE?	6
2.1	The problem of the independent variable	6
2.1.1	What differentiation actually does	6
2.1.2	Why this is not enough for consciousness	6
2.1.3	Three notions of time, sharply distinguished	6
2.2	The central claim and the hard problem	7
3	PART II — DIFFERENTIAL CALCULUS FROM THE GROUND UP	8
3.1	Functions, change and slope	8
3.1.1	Functions	8
3.1.2	Average rate of change	8
3.2	The limit	8
3.2.1	The idea	8
3.2.2	Formal definition (epsilon–delta)	8
3.2.3	Rules for limits	8
3.2.4	A fundamental limit: $\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1$	9
3.3	The derivative	9
3.3.1	Definition	9
3.3.2	The derivative from the definition: three worked cases	9
3.3.3	Differentiability implies continuity	9
3.4	All the differentiation rules	10
3.4.1	Rule 1 — The constant rule	10
3.4.2	Rule 2 — The power rule	10
3.4.3	Rule 3 — The constant multiple rule	10
3.4.4	Rule 4 — The sum and difference rule	10
3.4.5	Rule 5 — The product rule	10
3.4.6	Rule 6 — The quotient rule	11
3.4.7	Rule 7 — The chain rule	11
3.4.8	Rule 8 — Derivatives of the elementary functions	12
3.4.9	Rule 9 — Implicit differentiation	12
3.4.10	Rule 10 — Higher-order derivatives	13
3.4.11	Rule 11 — Partial derivatives	13

3.4.12	Rule 12 — The total derivative and the multivariable chain rule	13
3.5	Applications of the derivative	13
3.5.1	Tangent line and linear approximation	13
3.5.2	Extrema	13
3.5.3	Taylor series	14
4	PART III — INTEGRAL CALCULUS	15
4.1	The definite integral	15
4.1.1	Idea: accumulation	15
4.1.2	Definition via Riemann sums	15
4.2	The Fundamental Theorem of Calculus	15
4.3	The indefinite integral and standard antiderivatives	15
4.4	Integration techniques	16
4.4.1	Substitution (the chain rule in reverse)	16
4.4.2	Integration by parts (the product rule in reverse)	16
4.4.3	Numerical integration	16
4.5	Differential equations	17
4.5.1	Separation of variables	17
4.5.2	Numerical solution: Euler’s method	17
4.5.3	Reparametrising an ODE	17
5	PART IV — NONLINEAR DYNAMICS AND THE KURAMOTO MODEL	18
5.1	Dynamical systems and phase space	18
5.2	Fixed points and stability	18
5.3	Bifurcations	18
5.4	The logistic map and the route to chaos	19
5.5	Oscillators: phase, frequency, and the phase derivative	19
5.6	Two coupled oscillators	19
5.7	The Kuramoto model	20
5.7.1	The equation	20
5.7.2	The order parameter	20
5.7.3	The mean-field rewrite	20
5.7.4	The critical coupling	20
6	PART V — THE RTC REPARAMETRISATION: DIFFERENTIATING OVER THE PERCEPT	22
6.1	The transformation $dt = \tau(P) dP$	22
6.1.1	Motivation	22
6.1.2	Epistemic status of (5.1)	22
6.2	From $\frac{d\theta}{dt}$ to $\frac{d\theta}{dP}$	22
6.2.1	What changes mathematically, and what does not?	23
6.3	The Mind-Energy law $E_m = 1/\tau^3$	23
6.3.1	Step 1 — the kinetic-energy analogy	23
6.3.2	Step 2 — the temporal velocity (definitional)	23
6.3.3	Step 3 — substitution (inverse-square)	23
6.3.4	Step 4 — the effective mass (substantive RTC postulate)	24
6.3.5	Step 5 — combining	24
6.3.6	The scientifically defensible chain	24
6.3.7	The logarithmic form and the falsifiable exponent	24
6.4	The combined model: PPTQ as bridge	25
6.5	The conceptual hierarchy	25
7	PART VI — EPISTEMOLOGICAL ACCOUNTABILITY AND THE HARD PROBLEM	26

7.1	Four levels, sharply separated	26
7.2	The status of the mathematical layers (Chapter 9)	26
7.3	Why percept differentiation addresses the hard problem	26
7.4	Falsifiability: the concrete criteria	27
7.5	The epistemological transition summarised	27
8	PART VII — REFERENCE	28
8.1	Appendix A — All differentiation rules in one table	28
8.2	Appendix B — Standard integrals	28
8.3	Appendix C — List of symbols	28
8.4	Appendix D — Code examples	29
8.4.1	D.1 — Kuramoto (classical), R	29
8.4.2	D.2 — RTC Kuramoto (percept-indexed) with dynamic PPTQ, Python	29
8.4.3	D.3 — Mind-Energy law and log-log check, R	30
8.4.4	D.4 — Logistic map, iteration, R	30
8.5	Appendix E — Strogatz reading guide by topic	30
8.6	Appendix F — Cross-references to the thesis	31
9	Closing note	32

1 Preface: why this manual exists

This document is a self-contained mathematical companion to the doctoral thesis *The Resonating Self. A Resonance Theory of Consciousness* (RTC). In several places the thesis relies on differential and integral calculus and on nonlinear dynamics — most notably in Chapter 4 (the Psychophysical Time Quantum and Mind-Energy) and Chapter 9 (the Mathematical Formalism). For readers without an extensive mathematical background, and for examiners who wish to check the formal underpinnings step by step, a separate and thorough exposition is desirable. That exposition is given here.

The manual carries one guiding thread that distinguishes it from any standard calculus textbook. In ordinary analysis we differentiate a quantity *with respect to time*, and by “time” we almost always mean **homogeneous clock time** t , measured in seconds — the uniformly elapsing, directionless parameter of physics. In RTC the central discovery of the thesis is precisely the claim that the relevant independent variable for consciousness and human experience is *not* that homogeneous clock time, but **dynamic experiential time** τ , operationalised through the **percept** (the perceptual succession P) and the **Psychophysical Time Quantum** (PPTQ). Consciousness and experience must be differentiated with respect to the percept, not with respect to the second hand.

This manual takes the reader from the most elementary notions (change, slope, limit) to the point where this shift of independent variable can be made fully explicit and applied to the Kuramoto model. Along the way *all* the standard differentiation rules are covered and illustrated with numerous worked examples, followed by integral calculus and nonlinear dynamics.

1.1 The three books of Steven Strogatz as reference and source of inspiration

The applied mathematical style of the thesis — and of this manual — is deeply indebted to the work of **Steven H. Strogatz**. Three of his books are an explicit reference and source of inspiration:

1. *Nonlinear Dynamics and Chaos* (1994; 2nd ed. 2015). The standard textbook on dynamical systems. Strogatz treats *flows on the line*, fixed points and stability, bifurcations (saddle-node, transcritical, pitchfork, Hopf), the logistic map and the route to chaos through period doubling, and — in the chapter on coupled oscillators — the Kuramoto model. Part IV of this manual follows the structure and notation of this book. Its applied differential calculus (derivatives as velocity fields, linearisation through the Jacobian) is entirely in the service of the phenomenology of the system — exactly the attitude RTC adopts.
2. *Sync. The Emerging Science of Spontaneous Order* (2003). Strogatz’s popular-science account of synchronisation: fireflies flashing in unison, heart cells, applauding audiences, and the mathematical core of these — the Kuramoto model and the order parameter. This book supplies the conceptual motivation for the RTC claim that *coherence is a natural, unavoidable outcome* of coupled resonators, and not something that must be imposed from outside.
3. *Infinite Powers. How Calculus Reveals the Secrets of the Universe* (2019). Strogatz’s history and philosophy of calculus. The “Infinity Principle” (split a hard problem into infinitely many infinitely small, easy pieces and add them back up), the derivative as an instantaneous rate of change, the integral as accumulation, and the Fundamental Theorem that connects the two. Parts II and III of this manual follow the didactic line of this book: first *understand what change is*, only then calculate.

Wherever a concept is introduced for the first time in this manual, a boxed note (“**Strogatz reading guide**”) indicates which chapter in which book contains the extended treatment.

1.2 How to read this manual

- **Part I** lays down the guiding thread: the distinction between clock time t , experiential time τ and percept index P , and why this distinction bears on the *hard problem of consciousness*.
- **Part II** is a complete course in differential calculus: limit, derivative, and all the differentiation rules with proofs and worked examples.

- **Part III** is a course in integral calculus: the definite integral, the Fundamental Theorem, integration techniques, and the solution of differential equations (analytic and numerical).
- **Part IV** applies this to the nonlinear dynamics of Chapter 9: fixed points, bifurcations, the logistic map, oscillators and the Kuramoto model.
- **Part V** carries out the RTC reparametrisation: the transition $\frac{d\theta}{dt} \rightarrow \frac{d\theta}{dP}$ through $dt = \tau(P) dP$, the PPTQ-scaled Kuramoto model, and the derivation of the Mind-Energy law $E_m = 1/\tau^3$.
- **Part VI** discusses the epistemological status of all this: what the formalism does and does not prove, and how percept differentiation contributes to the RTC treatment of the hard problem.
- **Part VII** contains reference tables, a list of symbols and code examples in R and Python.

2 PART I — THE GUIDING THREAD: WITH RESPECT TO WHAT DO WE DIFFERENTIATE?

2.1 The problem of the independent variable

2.1.1 What differentiation actually does

Differentiation is the measurement of *change per unit of something else*. The derivative $\frac{df}{dx}$ answers the question: “If x increases by a minuscule amount, how much does f change, per unit of x ?” The choice of x — the **independent variable** — is therefore not a technical detail but a substantive decision about *which change we consider important*.

In physics that choice is almost always clock time t . The velocity of a particle is $\frac{dx}{dt}$; the acceleration is $\frac{d^2x}{dt^2}$; the phase change of an oscillator is $\frac{d\theta}{dt}$. Clock time is thereby taken to be:

- **homogeneous**: every second equals every other second;
- **directionless** (in dynamics): the parameter itself contains no “now”, no flow direction;
- **external**: the clock runs independently of the system being described.

2.1.2 Why this is not enough for consciousness

RTC claims that lived experience is not homogeneous, not directionless and not external. The experienced duration of an event depends on the state of consciousness: in deep meditation time “stretches”, in acute stress it “speeds up”, in flow it “disappears”. Two clock seconds can be phenomenologically utterly unequal. If we differentiate the dynamics of consciousness *with respect to clock time*, we average away precisely what we are trying to explain.

RTC resolves this by choosing a **different independent variable**. Experience unfolds in discrete chunks — **percepts** — each with a characteristic duration, the **PPTQ** τ (order of magnitude ~ 62 ms). The succession of percepts forms an ordered counting variable P : percept 1, percept 2, percept 3, The central RTC claim is:

The fundamental independent variable of conscious temporal dynamics is the perceptual succession P , scaled by the PPTQ τ , and not the external clock time t .

Mathematically this is captured by the relation

$$dt = \tau(P) dP \quad (1.1)$$

which says: one percept step ($dP = 1$) corresponds to a clock duration of $\tau(P)$ seconds, and τ may itself depend on the perceptual state P . Everything that follows in this manual is ultimately a tool for understanding, justifying and applying equation (1.1).

2.1.3 Three notions of time, sharply distinguished

Symbol	Name	Nature	Unit	Role in RTC
t	Clock time	homogeneous, continuous, external	second (s)	time of M_p (the brain domain), the physical description

Symbol	Name	Nature	Unit	Role in RTC
τ	PPTQ / experiential time quantum	state-dependent, “grainy”	seconds per percept (s)	duration of one perceptual moment; may vary with the state of consciousness
P	Perceptual succession / percept index	ordered, dimensionless, counts	— (dimensionless)	the actual independent variable of experience in M_e

The relation between t and P is *not* linear as soon as τ varies. That is precisely the mathematical content of the RTC discovery.

Strogatz reading guide. The idea that the choice of independent variable determines the physics of a problem runs as a theme through *Infinite Powers* (chapters 1–3: “calculus as the language of the universe”). The idea that time in a dynamical system is a *parameter* that may be rescaled is the core of the “flows” in *Nonlinear Dynamics and Chaos*, chapter 2.

2.2 The central claim and the hard problem

The *hard problem of consciousness* (Chalmers, 1995) is the question of why and how physical processes are accompanied by lived experience — why “there is something it is like” to undergo a particular brain process. The “easy” problems (attention, integration, reporting) can be described in functional or computational terms; the “hard” problem seems to escape this.

RTC does not claim that mathematics *explains away* the hard problem. The ontological core — that experience is primary (Axiom I) — is a metaphysical claim, not a claim that can be proven by an equation. What RTC does claim, and this is where this manual becomes relevant, is the following:

As soon as one differentiates the dynamics of consciousness with respect to the percept P rather than with respect to clock time t , a formal framework arises in which the temporal organisation of experience becomes an autonomous, measurable and falsifiable explanatory variable. The “explanatory gap” thereby shifts from an unbridgeable mystery to a *specific, quantitative model difference* between a t -indexed and a P -indexed description.

The burden of proof that the thesis carries is therefore not “here is a proof that qualia exist”, but: “here is a mathematically explicit, testable formulation in which the difference between physical time and experienced time does the work, with concrete predictions ($\beta = -3$ in the Mind-Energy law; $M_P \succ M_t$ in model comparison) that can fail.” Part VI develops this argument in full. First, however, the reader must master the mathematics.

3 PART II — DIFFERENTIAL CALCULUS FROM THE GROUND UP

3.1 Functions, change and slope

3.1.1 Functions

A **function** f assigns to each value of an input quantity x exactly one value $f(x)$ of an output quantity. Examples:

- $f(t) = 3t + 2$ — a straight line;
- $g(t) = t^2$ — a parabola;
- $\theta(t) = \omega t$ — the phase of an oscillator turning at constant angular frequency ω ;
- $\tau(P)$ — the PPTQ as a function of the percept index (in RTC not a constant).

3.1.2 Average rate of change

The **average rate of change** of f over the interval $[x, x + h]$ is

$$\frac{\Delta f}{\Delta x} = \frac{f(x + h) - f(x)}{h}. \quad (2.1)$$

Geometrically this is the slope of the **chord** (secant) through the points $(x, f(x))$ and $(x + h, f(x + h))$.

Worked example 2.1. Take $g(t) = t^2$, $x = 3$, $h = 0.1$:

$$\frac{g(3.1) - g(3)}{0.1} = \frac{9.61 - 9}{0.1} = \frac{0.61}{0.1} = 6.1.$$

With $h = 0.01$: $\frac{9.0601 - 9}{0.01} = 6.01$. With $h = 0.001$: 6.001. The values approach 6.

3.2 The limit

3.2.1 The idea

The **limit** $\lim_{h \rightarrow 0} F(h) = L$ means: by choosing h sufficiently close to 0 (but not equal to 0) one can bring $F(h)$ as close to L as desired. The limit asks about the behaviour *near* a point, not the value *at* that point — and that is exactly what is needed, because in (2.1) substituting $h = 0$ is forbidden (division by zero).

3.2.2 Formal definition (epsilon–delta)

$\lim_{h \rightarrow 0} F(h) = L$ if and only if: for every $\varepsilon > 0$ there exists a $\delta > 0$ such that $0 < |h| < \delta$ implies $|F(h) - L| < \varepsilon$.

3.2.3 Rules for limits

If $\lim F$ and $\lim G$ exist, then:

- $\lim(F \pm G) = \lim F \pm \lim G$;
- $\lim(F \cdot G) = \lim F \cdot \lim G$;
- $\lim(F/G) = \lim F / \lim G$ provided $\lim G \neq 0$;
- $\lim(c \cdot F) = c \cdot \lim F$.

Worked example 2.2. Compute $\lim_{h \rightarrow 0} \frac{(3 + h)^2 - 9}{h}$. Expand the numerator: $(9 + 6h + h^2) - 9 = 6h + h^2$. So

$$\frac{6h + h^2}{h} = 6 + h \quad (h \neq 0), \quad \lim_{h \rightarrow 0} (6 + h) = 6.$$

This confirms Worked example 2.1 exactly.

3.2.4 A fundamental limit: $\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1$

(with h in radians). This limit — provable geometrically via the squeeze theorem with $\cos h \leq \frac{\sin h}{h} \leq 1$ — is the engine behind the derivatives of $\sin$ and $\cos$, and hence behind all oscillator dynamics.

Strogatz reading guide. *Infinite Powers*, chapter 4 (“Putting Limits to Work”) treats precisely this limit and the “0/0” puzzle. The attitude “you may divide by h as long as $h \neq 0$, and only then take the limit” is the core of Strogatz’s Infinity Principle.

3.3 The derivative

3.3.1 Definition

The **derivative** of f at x is the limit of the chord slope as the interval shrinks to zero:

$$f'(x) = \frac{df}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}. \quad (2.2)$$

If this limit exists, f is **differentiable** at x . Geometrically $f'(x)$ is the slope of the **tangent line** to the graph at $(x, f(x))$; physically it is the **instantaneous rate of change**.

Notation. $f'(x)$ (Lagrange), $\frac{df}{dx}$ (Leibniz), $\dot{f}$ (Newton, for the derivative with respect to time). In this manual we use $\frac{d\theta}{dt}$ for the phase change with respect to clock time and $\frac{d\theta}{dP}$ for the phase change with respect to percept.

3.3.2 The derivative from the definition: three worked cases

Worked example 2.3 — $f(x) = x^2$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} = \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} = \lim_{h \rightarrow 0} (2x + h) = 2x.$$

Worked example 2.4 — $f(x) = x^3$. (Important for the Mind-Energy law.)

$$(x+h)^3 = x^3 + 3x^2h + 3xh^2 + h^3,$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 + h^3}{h} = \lim_{h \rightarrow 0} (3x^2 + 3xh + h^2) = 3x^2.$$

Worked example 2.5 — $f(x) = 1/x = x^{-1}$. (Important for $v_\tau = 1/\tau$.)

$$\frac{1}{x+h} - \frac{1}{x} = \frac{x - (x+h)}{x(x+h)} = \frac{-h}{x(x+h)},$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{-h}{hx(x+h)} = \lim_{h \rightarrow 0} \frac{-1}{x(x+h)} = -\frac{1}{x^2}.$$

3.3.3 Differentiability implies continuity

If f is differentiable at x , then f is continuous there. The converse does not hold: $f(x) = |x|$ is continuous at 0 but has a corner there and no derivative. This is relevant for RTC: at an *abrupt* state transition (for example a sudden jump in τ) the percept–time relation is continuous but not necessarily differentiable; one then works piecewise.

3.4 All the differentiation rules

Below are the rules one needs. Each rule is given with a brief justification and at least one worked example. A compact reference table appears at the end of the manual (Appendix A).

3.4.1 Rule 1 — The constant rule

If $f(x) = c$ (constant), then $f'(x) = 0$.

Reason: $f(x+h) - f(x) = c - c = 0$, so the difference quotient is 0.

Worked example. The intrinsic frequency ω_i of a Kuramoto oscillator is (per oscillator) constant: $\frac{d\omega_i}{dt} = 0$.

3.4.2 Rule 2 — The power rule

For every real exponent n : if $f(x) = x^n$, then

$$f'(x) = n x^{n-1}. \quad (2.3)$$

Reason (integer n): via the binomial expansion $(x+h)^n = x^n + nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \dots$; after dividing by h and letting $h \rightarrow 0$ only nx^{n-1} remains. For rational and real n the rule follows from the exponential/logarithmic rules (see Rule 8) together with the chain rule.

Worked examples.

$f(x)$	$f'(x)$
x^5	$5x^4$
x^{-3} (this is $1/x^3$)	$-3x^{-4} = -3/x^4$
$x^{1/2} = \sqrt{x}$	$\frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$
$x^{-1/3}$ (this is τ as a function of E_m , see Part V)	$-\frac{1}{3}x^{-4/3}$
x	1
$x^0 = 1$	0

3.4.3 Rule 3 — The constant multiple rule

If $g(x) = c f(x)$, then $g'(x) = c f'(x)$.

Worked example. $\frac{d}{d\tau} \left(\frac{1}{2} m \tau^{-2} \right) = \frac{1}{2} m \cdot (-2) \tau^{-3} = -m \tau^{-3}$. (This is the derivative of the kinetic-energy analogy with respect to τ ; see §12.4.)

3.4.4 Rule 4 — The sum and difference rule

$(f \pm g)'(x) = f'(x) \pm g'(x)$.

Worked example. $\frac{d}{dx} (3x^4 - 5x^2 + 7) = 12x^3 - 10x + 0 = 12x^3 - 10x$.

3.4.5 Rule 5 — The product rule

$$(f \cdot g)'(x) = f'(x) g(x) + f(x) g'(x). \quad (2.4)$$

Reason: write the numerator of the difference quotient as

$$f(x+h)g(x+h) - f(x)g(x) = \underbrace{[f(x+h) - f(x)] g(x+h)}_{\rightarrow f'(x)g(x)} + \underbrace{f(x) [g(x+h) - g(x)]}_{\rightarrow f(x)g'(x)},$$

divide by h and take the limit (using that g is continuous, so $g(x+h) \rightarrow g(x)$).

Worked example 2.6. $\frac{d}{dt}(t^2 \sin t) = 2t \sin t + t^2 \cos t$.

Worked example 2.7 (RTC-relevant). Suppose the PPTQ increases linearly with the percept, $\tau(P) = \tau_0(1 + aP)$, and the instantaneous phase change with respect to clock time $\frac{d\theta}{dt} = \omega$ is constant. Then the phase change with respect to percept is $\frac{d\theta}{dP} = \tau(P)\omega = \tau_0\omega(1 + aP)$, and the *acceleration in percept space* is

$$\frac{d}{dP} \left(\frac{d\theta}{dP} \right) = \frac{d}{dP} [\tau_0\omega + \tau_0\omega aP] = \tau_0\omega a.$$

A changing PPTQ therefore generates an apparent angular acceleration in the experiential dynamics, even though the underlying oscillator turns perfectly uniformly in clock time. This is one of the core mechanisms of RTC.

3.4.6 Rule 6 — The quotient rule

$$\left(\frac{f}{g} \right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}, \quad g(x) \neq 0. \quad (2.5)$$

Reason: write $f/g = f \cdot g^{-1}$ and combine the product and chain rules, or derive directly from the definition.

Worked example 2.8. $\frac{d}{d\tau} \left(\frac{1}{\tau^3} \right)$. Take $f = 1$, $g = \tau^3$: $f' = 0$, $g' = 3\tau^2$.

$$\frac{0 \cdot \tau^3 - 1 \cdot 3\tau^2}{\tau^6} = \frac{-3\tau^2}{\tau^6} = -\frac{3}{\tau^4}.$$

This is the derivative of the Mind-Energy law $E_m(\tau) = \tau^{-3}$ (identical to the power rule: $-3\tau^{-4}$). Interpretation: the shorter the PPTQ, the *faster* Mind-Energy rises upon further shortening — the sensitivity $\left| \frac{dE_m}{d\tau} \right| = 3/\tau^4$ grows very steeply.

Worked example 2.9. $\frac{d}{dx} \left(\frac{\sin x}{x} \right) = \frac{x \cos x - \sin x}{x^2}$.

3.4.7 Rule 7 — The chain rule

This is the most important rule for this manual, because the RTC reparametrisation is an application of it. If $y = f(u)$ and $u = g(x)$, then y is a function of x via the composition $y = f(g(x))$, and

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = f'(g(x)) \cdot g'(x). \quad (2.6)$$

Reason (sketch): $\frac{\Delta y}{\Delta x} = \frac{\Delta y}{\Delta u} \cdot \frac{\Delta u}{\Delta x}$ for $\Delta u \neq 0$; take the limit. The Leibniz notation makes the rule almost self-evident: the “ du ’s cancel” — a formally valid intuition as long as $g'(x) \neq 0$.

Worked example 2.10. $\frac{d}{dx} \sin(3x)$. Take $y = \sin u$, $u = 3x$: $\frac{dy}{du} = \cos u$, $\frac{du}{dx} = 3$. So $\frac{dy}{dx} = 3 \cos(3x)$.

Worked example 2.11. $\frac{d}{dt} (t^2 + 1)^{10} = 10(t^2 + 1)^9 \cdot 2t = 20t(t^2 + 1)^9$.

Worked example 2.12. $\frac{d}{dt} e^{-kt} = e^{-kt} \cdot (-k) = -k e^{-kt}$.

Worked example 2.13 — the RTC reparametrisation as a chain rule. Let $\theta = \theta(t)$ be the phase of an oscillator, and let clock time depend on the percept via $t = t(P)$ with $\frac{dt}{dP} = \tau(P)$ (this is $dt = \tau(P) dP$). Then the phase change *per percept* is:

$$\boxed{\frac{d\theta}{dP} = \frac{d\theta}{dt} \cdot \frac{dt}{dP} = \tau(P) \frac{d\theta}{dt}.} \quad (2.7)$$

This is the full mathematical content of the transition from the classical to the RTC Kuramoto model. See Part V for the elaboration and the numerical worked examples.

3.4.8 Rule 8 — Derivatives of the elementary functions

$f(x)$	$f'(x)$	Note
$\sin x$	$\cos x$	x in radians; rests on $\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1$
$\cos x$	$-\sin x$	
$\tan x$	$\sec^2 x = 1/\cos^2 x$	via quotient rule on $\sin/\cos$
e^x	e^x	the function that is its own derivative
a^x	$a^x \ln a$	
$\ln x$	$1/x$	$x > 0$
$\log_a x$	$1/(x \ln a)$	
$\arcsin x$	$1/\sqrt{1-x^2}$	relevant for phase locking, §11.2
$\arctan x$	$1/(1+x^2)$	

Derivation of $(\sin x)' = \cos x$:

$$\frac{\sin(x+h) - \sin x}{h} = \frac{\sin x \cos h + \cos x \sin h - \sin x}{h} = \sin x \cdot \frac{\cos h - 1}{h} + \cos x \cdot \frac{\sin h}{h}.$$

With $\frac{\sin h}{h} \rightarrow 1$ and $\frac{\cos h - 1}{h} \rightarrow 0$ this yields $\cos x$.

Worked example 2.14. $\frac{d}{dx} \ln(\cos x) = \frac{1}{\cos x} \cdot (-\sin x) = -\tan x$ (chain rule + Rule 8).

3.4.9 Rule 9 — Implicit differentiation

Sometimes y is not given explicitly as $y = f(x)$, but implicitly through an equation $F(x, y) = 0$. One then differentiates both sides with respect to x , treating y as a function of x (chain rule), and solves for $\frac{dy}{dx}$.

Worked example 2.15. The order-parameter relation $R^2 = \left(\frac{1}{N} \sum \cos \theta_j\right)^2 + \left(\frac{1}{N} \sum \sin \theta_j\right)^2$. To find how R changes when all phases shift by δ , one differentiates implicitly; the result is $\frac{dR}{d\delta} = 0$ (a global phase shift does not change the coherence — the rotational invariance of the Kuramoto model).

Worked example 2.16. From $\log E_m = -3 \log \tau$ implicit differentiation with respect to τ gives $\frac{1}{E_m} \frac{dE_m}{d\tau} = -\frac{3}{\tau}$, so $\frac{dE_m}{d\tau} = -\frac{3E_m}{\tau} = -\frac{3}{\tau^4}$ — consistent with Worked example 2.8. The quantity $\frac{d \ln E_m}{d \ln \tau} = -3$ is the **elasticity**, and this number -3 is precisely the falsifiable prediction (see §12.5 and Part VI).

3.4.10 Rule 10 — Higher-order derivatives

The second derivative $f''(x) = \frac{d}{dx}(f'(x)) = \frac{d^2 f}{dx^2}$ measures the *change of the rate of change* (curvature; physically: acceleration). Analogously $f^{(n)}$.

Worked example. $\theta(t) = \sin(\omega t)$: $\theta'(t) = \omega \cos(\omega t)$, $\theta''(t) = -\omega^2 \sin(\omega t) = -\omega^2 \theta$. This is the harmonic-oscillator equation $\ddot{\theta} + \omega^2 \theta = 0$.

3.4.11 Rule 11 — Partial derivatives

For a function of several variables, $f(x_1, \dots, x_n)$, the **partial derivative** $\frac{\partial f}{\partial x_k}$ is the derivative with respect to x_k while the other variables are held constant.

Worked example 2.17 — the Kuramoto coupling term. Consider, for oscillator i , the function

$$H_i(\theta_1, \dots, \theta_N) = \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i).$$

The partial derivatives are:

$$\frac{\partial H_i}{\partial \theta_i} = -\frac{K}{N} \sum_{j=1}^N \cos(\theta_j - \theta_i), \quad \frac{\partial H_i}{\partial \theta_k} = \frac{K}{N} \cos(\theta_k - \theta_i) \quad (k \neq i).$$

These partial derivatives form the **Jacobian** of the Kuramoto system, which one needs to determine the stability of the synchronised state (§11.4).

3.4.12 Rule 12 — The total derivative and the multivariable chain rule

If $f = f(u, v)$ with $u = u(t)$, $v = v(t)$, then

$$\frac{df}{dt} = \frac{\partial f}{\partial u} \frac{du}{dt} + \frac{\partial f}{\partial v} \frac{dv}{dt}. \quad (2.8)$$

This is the generalisation of Rule 7 needed when τ depends both explicitly on P and through other state variables — for example $\tau = \tau(P, E_m(P))$.

3.5 Applications of the derivative

3.5.1 Tangent line and linear approximation

The **linear approximation** (first-order Taylor) of f around x_0 :

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0). \quad (2.9)$$

This is the tool with which one *linearises* a nonlinear system around a fixed point (§10.3). Strogatz calls this “the derivative as a local, linear approximation” (*Infinite Powers*, ch. 6).

Worked example 2.18. $\sqrt{101} \approx \sqrt{100} + \frac{1}{2\sqrt{100}}(1) = 10 + 0.05 = 10.05$ (exact: 10.0499...).

3.5.2 Extrema

At a local maximum or minimum of a differentiable function, $f'(x) = 0$ (stationary point). The sign of $f''(x)$ distinguishes a maximum ($f'' < 0$) from a minimum ($f'' > 0$).

Worked example 2.19. The Kuramoto coupling force on oscillator i from one neighbour, $g(\phi) = K \sin \phi$ with $\phi = \theta_j - \theta_i$, is maximal where $g'(\phi) = K \cos \phi = 0$, hence at $\phi = \pi/2$: a phase difference of 90° produces the strongest “pull” toward synchronisation.

3.5.3 Taylor series

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \frac{f'''(x_0)}{3!}(x - x_0)^3 + \dots \quad (2.10)$$

Worked example 2.20. $\sin \phi = \phi - \frac{\phi^3}{6} + \dots$, so for small phase differences $\sin \phi \approx \phi$. This approximation turns the Kuramoto model in the weak-coupling limit into a *linear* system — exactly how Strogatz derives the synchronisation threshold analytically (*Nonlinear Dynamics and Chaos*, §8.6; *Sync*, ch. 2).

Strogatz reading guide. *Infinite Powers* ch. 6–7 (derivative, extrema, approximations); *Nonlinear Dynamics and Chaos* §2.4 (linear stability analysis) and §8.6 (the Kuramoto model).

4 PART III — INTEGRAL CALCULUS

4.1 The definite integral

4.1.1 Idea: accumulation

Where the derivative *divides* (change per unit), the integral *adds up* (accumulation over an interval). If $\frac{d\theta}{dt} = \omega(t)$ is the instantaneous angular frequency, then the *total accumulated phase* between $t = a$ and $t = b$ is:

$$\theta(b) - \theta(a) = \int_a^b \omega(t) dt.$$

4.1.2 Definition via Riemann sums

Divide $[a, b]$ into n pieces of width $\Delta t = (b-a)/n$, with sample points t_k^* . The **Riemann sum** is $\sum_{k=1}^n f(t_k^*) \Delta t$.

The **definite integral** is the limit:

$$\int_a^b f(t) dt = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(t_k^*) \Delta t. \quad (3.1)$$

Geometrically: the net area between the graph and the t -axis (area below the axis counts negatively).

Strogatz reading guide. This *is* the Infinity Principle of *Infinite Powers* (ch. 1–3, 8): cut the region into infinitely many infinitely thin strips and add them up.

Worked example 3.1. $\int_0^3 2t dt$. Analytically (see Fundamental Theorem): $[t^2]_0^3 = 9$. Numerically with $n = 3$ ($\Delta t = 1$, right sample points $t = 1, 2, 3$): $2(1) + 2(2) + 2(3) = 2 + 4 + 6 = 12$; with $n = 6$: approaches 9; with $n \rightarrow \infty$: exactly 9.

4.2 The Fundamental Theorem of Calculus

This theorem connects differentiation and integration as inverses of each other.

Part 1. If $F(x) = \int_a^x f(t) dt$, then $F'(x) = f(x)$.

Part 2. If F is an antiderivative of f (that is, $F' = f$), then

$$\int_a^b f(t) dt = F(b) - F(a). \quad (3.2)$$

In practice: to integrate, find a function whose derivative is the integrand.

Worked example 3.2. $\int_a^b \omega dt = [\omega t]_a^b = \omega(b-a)$. At constant angular frequency the accumulated phase is simply $\omega \cdot$ elapsed time.

4.3 The indefinite integral and standard antiderivatives

$\int f(x) dx = F(x) + C$ with $F' = f$ and C an arbitrary constant of integration.

$f(x)$	$\int f(x) dx$
$x^n \ (n \neq -1)$	$\frac{x^{n+1}}{n+1} + C$
$x^{-1} = 1/x$	$\ln x + C$
e^{kx}	$\frac{1}{k}e^{kx} + C$
$\sin(kx)$	$-\frac{1}{k}\cos(kx) + C$
$\cos(kx)$	$\frac{1}{k}\sin(kx) + C$
$\frac{1}{1+x^2}$	$\arctan x + C$
$\frac{1}{\sqrt{1-x^2}}$	$\arcsin x + C$

Worked example 3.3. $\int (12t^3 - 10t + 7) dt = 3t^4 - 5t^2 + 7t + C$ (check by differentiating: Worked example 2.6 style).

4.4 Integration techniques

4.4.1 Substitution (the chain rule in reverse)

$$\int f(g(x)) g'(x) dx = \int f(u) du, \quad u = g(x), \quad du = g'(x) dx. \quad (3.3)$$

Worked example 3.4. $\int 2t(t^2 + 1)^{10} dt$. Take $u = t^2 + 1$, $du = 2t dt$: $\int u^{10} du = \frac{u^{11}}{11} + C = \frac{(t^2 + 1)^{11}}{11} + C$.

Worked example 3.5 — the clock-time-percept-time transformation as substitution. The accumulated phase from percept $P = 0$ to $P = P_1$ is

$$\theta(P_1) - \theta(0) = \int_0^{P_1} \frac{d\theta}{dP} dP = \int_0^{P_1} \tau(P) \frac{d\theta}{dt} dP.$$

With the substitution $dt = \tau(P) dP$ this passes into $\int_{t(0)}^{t(P_1)} \frac{d\theta}{dt} dt$ — the ordinary clock integral. The two descriptions therefore give *the same* total phase; they differ in *how that phase is distributed over the axis*. That difference in distribution is what RTC considers empirically testable (Part VI).

4.4.2 Integration by parts (the product rule in reverse)

$$\int u dv = uv - \int v du. \quad (3.4)$$

Worked example 3.6. $\int t \sin t dt$. Take $u = t$ ($du = dt$), $dv = \sin t dt$ ($v = -\cos t$): $-t \cos t + \int \cos t dt = -t \cos t + \sin t + C$.

4.4.3 Numerical integration

When no antiderivative is known: the **trapezoidal rule** $\int_a^b f \approx \Delta t [\frac{1}{2}f_0 + f_1 + \dots + f_{n-1} + \frac{1}{2}f_n]$, or **Simpson's rule**. This is also exactly how one solves differential equations numerically (§3.6).

Worked example 3.7 — proper-time analogy (from the RTC relativity module). In the RTC Lorentz analogy the integrated “proper time” is $\tau_p = \int_0^T \sqrt{1 - v(t)^2} dt$ with $v(t)$ the relative information speed (fraction of Ω_{pptq}). For $v(t) = 0.5 + 0.4 \sin t$ over $[0, 10]$ with step $\Delta t = 0.1$ the trapezoidal rule gives $\tau_p \approx 7.7$ (against $T = 10$ clock seconds): the subjective proper time is shorter than the clock time as the “speed” rises.

4.5 Differential equations

An **ordinary differential equation** (ODE) relates an unknown function to its derivatives, for example $\frac{d\theta}{dt} = \omega + K \sin(\Psi - \theta)$. To *solve* an ODE means: to find the function $\theta(t)$ that satisfies it.

4.5.1 Separation of variables

If $\frac{dy}{dx} = A(x) B(y)$, then $\int \frac{dy}{B(y)} = \int A(x) dx$.

Worked example 3.8 — the Adler equation (two coupled oscillators). The phase difference $\phi = \theta_2 - \theta_1$ of two oscillators satisfies (see §11.2)

$$\frac{d\phi}{dt} = \Delta\omega - K \sin \phi.$$

In a locked state $\frac{d\phi}{dt} = 0$, so $\sin \phi^* = \frac{\Delta\omega}{K}$. This has a solution (and hence phase locking) if and only if $|\Delta\omega| \leq K$. This is Strogatz’s classic result on the synchronisation threshold (*Nonlinear Dynamics and Chaos*, §4.5 and §8.2; *Sync*, ch. 2).

4.5.2 Numerical solution: Euler’s method

$$\theta(t + \Delta t) \approx \theta(t) + \Delta t \cdot \frac{d\theta}{dt}. \quad (3.5)$$

This is nothing but the linear approximation (2.9) applied repeatedly. For better accuracy one uses **Runge–Kutta 4 (RK4)**.

Worked example 3.9 — Euler on one oscillator. $\frac{d\theta}{dt} = \omega$ with $\omega = 2\pi \cdot 10 \approx 62.83$ rad/s (a 10 Hz alpha oscillator), $\theta(0) = 0$, $\Delta t = 0.01$ s. After one step: $\theta = 0.628$ rad. After 6.2 steps ($t = 0.062$ s = τ): $\theta \approx 3.90$ rad $\approx 223^\circ$. Per PPTQ window the oscillator therefore turns roughly $\frac{2}{3}$ of a full cycle.

4.5.3 Reparametrising an ODE

Given $\frac{d\theta}{dt} = f(\theta)$ and $\frac{dt}{dP} = \tau(P)$, the chain rule gives directly the reparametrised ODE:

$$\frac{d\theta}{dP} = \tau(P) f(\theta). \quad (3.6)$$

This is the formal basis of the PPTQ-scaled Kuramoto model (Part V). Note: the *trajectory* (the path θ takes) remains the same set of points; only the *speed at which the path is traversed* changes, and with it the relation to the external clock.

Strogatz reading guide. *Nonlinear Dynamics and Chaos* ch. 2 (flows, solutions as trajectories), appendix on numerical methods; *Infinite Powers* ch. 8–10 (integral, Fundamental Theorem, differential equations as “the operating system of the universe”).

5 PART IV — NONLINEAR DYNAMICS AND THE KURAMOTO MODEL

5.1 Dynamical systems and phase space

A **dynamical system** is given by a state vector $\mathbf{x}(t) \in \mathbb{R}^n$ and a *vector field* $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$ that at each point prescribes the instantaneous speed and direction. The **phase space** is the space of all possible states; a **trajectory** is a solution curve.

Examples from Chapter 9 of the thesis:

- **Logistic map** (discrete): $x_{n+1} = r x_n (1 - x_n)$ — model for reflexive amplification and the route to chaos.
- **Kuramoto model** (continuous, N -dimensional): $\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_j \sin(\theta_j - \theta_i)$ — model for percept formation through synchronisation.
- **Ginzburg–Landau field** (continuum field): $\partial_t \psi = \psi - (1 + i\omega)|\psi|^2 \psi + \nabla^2 \psi + \eta$ — heuristic description of the holographic field.

5.2 Fixed points and stability

A **fixed point** (equilibrium) $\mathbf{x}^*$ satisfies $\mathbf{F}(\mathbf{x}^*) = \mathbf{0}$: the system stands still there. **Stability** is determined by linearisation: write $\mathbf{x} = \mathbf{x}^* + \delta$, then $\dot{\delta} \approx J(\mathbf{x}^*) \delta$ with J the **Jacobian** (matrix of partial derivatives, Rule 11). The signs of the real parts of the eigenvalues of J determine whether perturbations grow (unstable) or decay (stable).

Worked example 4.1 — 1D. $\dot{x} = f(x) = rx - x^3$ (pitchfork normal form). Fixed points: $x^* = 0$ and (for $r > 0$) $x^* = \pm\sqrt{r}$. Linearisation: $f'(x) = r - 3x^2$. At $x^* = 0$: $f'(0) = r$ — stable for $r < 0$, unstable for $r > 0$. At $x^* = \pm\sqrt{r}$: $f'(\pm\sqrt{r}) = r - 3r = -2r < 0$ — stable. At $r = 0$ the number and nature of the equilibria change: a **bifurcation**.

5.3 Bifurcations

A **bifurcation** is a qualitative change in the dynamics when a control parameter crosses a critical value. The four standard types (Strogatz, *Nonlinear Dynamics and Chaos*, ch. 3 and 8):

Type	Normal form	What happens	RTC interpretation (Ch. 9/11)
Saddle-node	$\dot{x} = r + x^2$	two equilibria appear/disappear	sudden emergence/disappearance of a percept state
Transcritical	$\dot{x} = rx - x^2$	two equilibria exchange stability	dominance switch $M_p \leftrightarrow M_e$
Pitchfork	$\dot{x} = rx - x^3$	one equilibrium splits into three	first bifurcation: emergence of bimodal consciousness ($r \approx 2.9$)
Hopf	$\dot{r} = \mu r - r^3, \dot{\varphi} = \omega$	equilibrium becomes a limit cycle	transition to mood cycles (bipolar disorder); parameter γ = reflexive gain

Worked example 4.2 — saddle-node. $\dot{x} = r + x^2$. For $r < 0$: $x^* = \pm\sqrt{-r}$ (two equilibria, one stable, one unstable). For $r = 0$: one (semi-stable) equilibrium. For $r > 0$: no equilibria — the system “escapes”. The critical value is $r_c = 0$.

5.4 The logistic map and the route to chaos

$$x_{n+1} = r x_n (1 - x_n), \quad x_n \in [0, 1], \quad r \in [0, 4]. \quad (4.1)$$

This is equation (9.6) from the thesis. The derivative of the map is $f'(x) = r(1 - 2x)$.

Fixed point. $x^* = r x^* (1 - x^*) \Rightarrow x^* = 0$ or $x^* = 1 - \frac{1}{r}$.

Stability of $x^* = 1 - 1/r$. $f'(x^*) = r(1 - 2(1 - \frac{1}{r})) = r(-1 + \frac{2}{r}) = 2 - r$. Stable as long as $|2 - r| < 1$, hence $1 < r < 3$. At $r = 3$, $|f'| = 1$: **period doubling** to a 2-cycle.

Worked example 4.3 — iteration at $r = 3.2$, $x_0 = 0.5$.

n	x_n
0	0.5000
1	$3.2 \cdot 0.5 \cdot 0.5 = 0.8000$
2	$3.2 \cdot 0.8 \cdot 0.2 = 0.5120$
3	$3.2 \cdot 0.512 \cdot 0.488 = 0.7995$
4	0.5130
5	0.7994

The system settles into a **2-cycle** alternating between ≈ 0.513 and ≈ 0.799 . The (formerly stable) fixed point $x^* = 1 - 1/3.2 = 0.6875$ is now unstable ($f'(x^*) = 2 - 3.2 = -1.2$, $|-1.2| > 1$).

The period-doubling cascade. Further increase of r gives 4-, 8-, 16-cycles over ever shorter parameter intervals, with the universal **Feigenbaum constant** $\delta \approx 4.669$ as the convergence ratio. At $r_\infty \approx 3.5699$ **chaos** begins: sensitive dependence on initial conditions. This is Phase III in the thesis (“chaotic de-resonance”, $r > 3.57$).

Strogatz reading guide. *Nonlinear Dynamics and Chaos* ch. 10 (“One-Dimensional Maps”) treats the logistic map, the cascade and Feigenbaum universality in full.

5.5 Oscillators: phase, frequency, and the phase derivative

An **oscillator** is a system that runs through a cycle. Its state is summarised by a single angle, the **phase** $\theta \in [0, 2\pi)$. The **angular frequency** is the phase derivative:

$$\omega = \frac{d\theta}{dt}.$$

A *uniform* oscillator has $\frac{d\theta}{dt} = \omega$ constant, so $\theta(t) = \theta_0 + \omega t$ (integration: Worked example 3.2). The period is $T = 2\pi/\omega$; the ordinary frequency $f = \omega/2\pi$ in Hz.

Worked example 4.4. A gamma oscillator of 40 Hz: $\omega = 2\pi \cdot 40 \approx 251.3$ rad/s, $T = 25$ ms. Within one PPTQ window of 62 ms it completes $62/25 \approx 2.5$ cycles.

5.6 Two coupled oscillators

Let θ_1, θ_2 be two oscillators with intrinsic frequencies ω_1, ω_2 , sinusoidally coupled with strength K :

$$\dot{\theta}_1 = \omega_1 + K \sin(\theta_2 - \theta_1), \quad \dot{\theta}_2 = \omega_2 + K \sin(\theta_1 - \theta_2).$$

Define the phase difference $\phi = \theta_2 - \theta_1$. Then (sum and difference rule, plus $\sin(-\phi) = -\sin \phi$):

$$\dot{\phi} = \dot{\theta}_2 - \dot{\theta}_1 = (\omega_2 - \omega_1) - 2K \sin \phi = \Delta\omega - 2K \sin \phi. \quad (4.2)$$

This is the **Adler equation**. Phase locking ($\dot{\phi} = 0$) requires $\sin \phi^* = \frac{\Delta\omega}{2K}$, solvable if and only if $|\Delta\omega| \leq 2K$.

Worked example 4.5. $\omega_1 = 2\pi(10)$, $\omega_2 = 2\pi(11)$ rad/s, so $\Delta\omega = 2\pi \approx 6.28$ rad/s. Locking requires $K \geq \Delta\omega/2 \approx 3.14$ rad/s. At $K = 4$: $\sin \phi^* = 6.28/8 = 0.785$, $\phi^* = \arcsin(0.785) \approx 0.903$ rad $\approx 51.7^\circ$ — the oscillators turn locked, with the faster one 51.7° ahead.

5.7 The Kuramoto model

5.7.1 The equation

For N oscillators with frequencies ω_i drawn from a distribution $g(\omega)$, each coupled to all the others:

$$\frac{d\theta_i}{dt} = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i), \quad i = 1, \dots, N. \quad (4.3)$$

This is equation (9.3) from the thesis (without the noise term ξ_i). The factor $1/N$ makes the coupling *intensive* (independent of the system size).

5.7.2 The order parameter

Define the complex number

$$R e^{i\Psi} = \frac{1}{N} \sum_{j=1}^N e^{i\theta_j}. \quad (4.4)$$

Here $R \in [0, 1]$ is the **coherence** ($0 =$ fully incoherent, $1 =$ perfect synchronisation) and Ψ the **mean phase**. This is equation (9.4). Geometrically: place each oscillator as a unit vector on the unit circle; R is the length of their average (“the centre of mass”).

5.7.3 The mean-field rewrite

Multiply (4.4) by $e^{-i\theta_i}$ and take the imaginary part:

$$\frac{1}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i) = R \sin(\Psi - \theta_i).$$

Substitution into (4.3) gives the elegant **mean-field form**:

$$\frac{d\theta_i}{dt} = \omega_i + KR \sin(\Psi - \theta_i). \quad (4.5)$$

Each oscillator feels the population *only* through the mean field (R, Ψ) . The effective coupling is KR : the more coherent the population, the more strongly it pulls each oscillator toward itself — a positive feedback that makes synchronisation **self-reinforcing**.

5.7.4 The critical coupling

For a symmetric, unimodal $g(\omega)$ with peak at ω_0 , the self-consistency analysis (Strogatz, 2000) shows that there is a threshold:

$$K_c = \frac{2}{\pi g(\omega_0)}. \quad (4.6)$$

For $K < K_c$: $R = 0$ (incoherence). For $K > K_c$: R grows continuously from 0 as $R \propto \sqrt{K - K_c}$ — a second-order phase transition. In the thesis $K_c \approx 1.3$ is used (Figure 15), with the percept threshold $R_c \approx 0.6$.

Worked example 4.6 — Euler simulation of (4.3), outline. Take $N = 50$, ω_i from a normal distribution (sd = 0.5), $\Delta t = 0.01$. Per step: (1) compute R, Ψ from (4.4); (2) update each θ_i with $\theta_i \leftarrow \theta_i + \Delta t [\omega_i +$

$KR \sin(\Psi - \theta_i)]$. At $K = 0.8$, R stays ≈ 0.15 (noise around zero); at $K = 2.0$, R rises within a few hundred steps to ≈ 0.8 . This reproduces Figure 15 of the thesis. Full R and Python code is in Appendix D.

Strogatz reading guide. *Sync* ch. 1–3 (the story, the fireflies, the order parameter); *Nonlinear Dynamics and Chaos* §8.6 (2nd ed.: full mathematical treatment of (4.3)–(4.6)); Strogatz (2000), *Physica D* 143, 1–20 (“From Kuramoto to Crawford”) — the review article the thesis cites.

6 PART V — THE RTC REPARAMETRISATION: DIFFERENTIATING OVER THE PERCEPT

Now all the threads come together. This part carries out, step by step, the transition from the classical, clock-indexed Kuramoto model to the PPTQ-scaled, percept-indexed RTC model, and in addition derives the Mind-Energy law $E_m = 1/\tau^3$ in full.

6.1 The transformation $dt = \tau(P) dP$

6.1.1 Motivation

In the classical model t (seconds) is the independent variable. RTC claims: conscious experience unfolds in **percepts**, discrete integration windows. Number them $P = 1, 2, 3, \dots$ Each percept spans a clock duration equal to the PPTQ τ of that moment. The infinitesimal version of this is:

$$\boxed{dt = \tau(P) dP} \quad (5.1)$$

$\tau(P)$ is **dynamic**: it depends on the state of consciousness, which itself evolves with P . The empirically expected ordering is

$$\tau_{\text{meditation}} > \tau_{\text{waking}} > \tau_{\text{stress}}. \quad (5.2)$$

6.1.2 Epistemic status of (5.1)

As the project notes emphasise: (5.1) is *not* a pure mathematical identity unless P is explicitly defined as a **dimensionless percept-count variable**. We do so here: P counts percepts, $[P] = 1$, and τ carries the unit second. Under that definition (5.1) is a *modelling assumption with content*: it claims that the natural “tick” of conscious dynamics is the percept, not the second. Precisely because it is an assumption with content, it is testable (Part VI).

6.2 From $\frac{d\theta}{dt}$ to $\frac{d\theta}{dP}$

Apply the chain rule (Rule 7, equation 2.7):

$$\frac{d\theta_i}{dP} = \frac{d\theta_i}{dt} \cdot \frac{dt}{dP} = \tau(P) \frac{d\theta_i}{dt}. \quad (5.3)$$

Substitute the classical Kuramoto right-hand side (4.3):

$$\boxed{\frac{d\theta_i}{dP} = \tau(P) \left[\omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j(P) - \theta_i(P)) \right]}. \quad (5.4)$$

This is the **RTC Kuramoto model** — equation (9.x) in the revised Chapter 9, §9.4. In mean-field form:

$$\frac{d\theta_i}{dP} = \tau(P) [\omega_i + K R(P) \sin(\Psi(P) - \theta_i)], \quad (5.5)$$

with the percept-indexed order parameter

$$R(P) e^{i\Psi(P)} = \frac{1}{N} \sum_{j=1}^N e^{i\theta_j(P)}. \quad (5.6)$$

6.2.1 What changes mathematically, and what does not?

- The **set of trajectories** in phase space remains identical: (5.4) traverses the same curves as (4.3), because the factor $\tau(P) > 0$ rescales only the speed, not the direction of the vector field (§3.6).
- The **relation to the external clock** does change: where τ is large (meditation), *less* perceptual change happens per clock second; where τ is small (stress), *more*.
- The **coherence build-up per percept** becomes: $\frac{dR}{dP} = \tau(P) \cdot \frac{dR}{dt}$. At short τ the system accumulates less coherence per percept — a concrete prediction about how percept formation degrades under stress.

Worked example 5.1 — one oscillator, constant τ . $\frac{d\theta}{dt} = \omega = 2\pi(10)$ rad/s, $\tau = 0.062$ s. Then $\frac{d\theta}{dP} = \tau\omega = 0.062 \cdot 62.83 \approx 3.90$ rad per percept. Integrating over P (Worked example 3.2 style): $\theta(P) = 3.90 P$. After 10 percepts: $\theta = 39.0$ rad ≈ 6.2 full revolutions. In clock time: 10 percepts = 0.62 s, and $\omega t = 62.83 \cdot 0.62 = 38.96$ rad — consistent (the small difference is rounding).

Worked example 5.2 — dynamic $\tau(P)$. Let $\tau(P) = 0.062(1 - 0.3e^{-P/5})$ model the PPTQ recovering from a shortened (stress) initial state $\tau(0) = 0.0434$ s toward 0.062 s. With $\omega = 62.83$ rad/s:

P	$\tau(P)$ (s)	$d\theta/dP = \tau\omega$ (rad)
0	0.0434	2.73
2	0.0495	3.11
5	0.0552	3.47
10	0.0595	3.74
∞	0.0620	3.95

The accumulated phase after P_1 percepts is $\theta(P_1) = \int_0^{P_1} \tau(P)\omega dP = \omega \left[0.062 P + 0.062 \cdot 0.3 \cdot 5 e^{-P/5} \right]_0^{P_1}$ (standard integral $\int e^{-P/5} dP = -5e^{-P/5}$). For $P_1 = 10$: $\theta(10) = 62.83(0.062 \cdot 10 + 0.093(e^{-2} - 1)) = 62.83(0.620 - 0.0804) = 62.83 \cdot 0.5396 \approx 33.9$ rad. Note that this is *less* than the 39.0 rad from Worked example 5.1: the shortened PPTQ in the initial state “costs” accumulated phase.

6.3 The Mind-Energy law $E_m = 1/\tau^3$

6.3.1 Step 1 — the kinetic-energy analogy

The thesis (§4.11) postulates a PPTQ-dependent Mind-Energy by analogy with the classical (Newtonian) kinetic energy:

$$M_E = \frac{1}{2} m v_\tau^2, \quad (5.7)$$

where v_τ is a *temporal velocity* associated with the PPTQ.

6.3.2 Step 2 — the temporal velocity (definitional)

The natural RTC temporal velocity is the reciprocal PPTQ:

$$v_\tau = \frac{1}{\tau} = \tau^{-1}. \quad (5.8)$$

Interpretation: τ is the duration of one psychophysical cycle, so $1/\tau$ is the *cycle frequency* — the shorter the PPTQ, the faster the conscious “refresh”. This is a **definition**, not an assumption.

6.3.3 Step 3 — substitution (inverse-square)

Substituting (5.8) into (5.7):

$$M_E = \frac{1}{2} m \left(\frac{1}{\tau} \right)^2 = \frac{m}{2\tau^2} \propto \tau^{-2}. \quad (5.9)$$

At this point we have an *inverse-square* law, not yet the inverse-cubic one.

6.3.4 Step 4 — the effective mass (substantive RTC postulate)

To obtain the inverse-cubic law, the effective psychophysical mass must also scale inversely with the PPTQ:

$$m_\tau \propto \frac{1}{\tau} = \tau^{-1}. \quad (5.10)$$

This is **not** a consequence of classical mechanics; it is a *substantive RTC assumption* that requires independent theoretical justification *and* empirical testing. Honestly naming which part is inherited analogy and which part is new physics/metaphysics strengthens the argument.

6.3.5 Step 5 — combining

$$E_m \propto m_\tau v_\tau^2 \propto \tau^{-1} \cdot \tau^{-2} = \tau^{-3}.$$

$$\boxed{E_m = \frac{1}{\tau^3}} \quad (5.11)$$

(with the factor $\frac{1}{2}$ absorbed into the normalisation of the Mind-Energy unit). The inverse-cubic law therefore arises from **two independent temporal scalings**: one in the velocity (τ^{-2}) and one in the mass (τ^{-1}).

6.3.6 The scientifically defensible chain

$$\text{kinetic-energy analogy} \rightarrow v_\tau = \tau^{-1} \rightarrow m_\tau \propto \tau^{-1} \rightarrow E_m \propto \tau^{-3},$$

with the **crucial caveat** that $m_\tau \propto \tau^{-1}$ is an additional postulate. Note: $\frac{1}{2}mv^2$ is the Newtonian low-velocity expression, not the relativistic $E = \gamma mc^2$. The thesis therefore speaks of a *kinetic-energy analogy* or *relativistically inspired analogy*, not of a derivation from special relativity.

6.3.7 The logarithmic form and the falsifiable exponent

Take the logarithm of (5.11):

$$\ln E_m = -3 \ln \tau. \quad (5.12)$$

A log-log regression of independently measured E_m against independently measured τ must yield a slope

$$\boxed{\beta = -3.}$$

An estimated $\beta \approx -2$ would indicate an incorrect mass postulate (only the velocity scaling survives); $\beta \approx -1$ or $\beta > 0$ would refute the law. This is the direct statistical test criterion (see §12.5 and Part VI).

Worked example 5.3 — numerical values of E_m . With $E_m = 1/\tau^3$ (τ in seconds):

State	τ	$E_m = \tau^{-3}$	$\ln \tau$	$\ln E_m$
Deep meditation	0.100 s	1.00×10^3	-2.303	6.908
Relaxed waking	0.080 s	1.95×10^3	-2.526	7.577
Normal waking	0.062 s	4.20×10^3	-2.781	8.342
Heightened alertness	0.050 s	8.00×10^3	-2.996	8.987
Acute stress	0.042 s	1.35×10^4	-3.170	9.510

Check of the slope between the first and last rows: $\frac{9.510 - 6.908}{-3.170 - (-2.303)} = \frac{2.602}{-0.867} = -3.00$. Exactly $\beta = -3$, as (5.12) requires.

Worked example 5.4 — the sensitivity $dE_m/d\tau$. From Worked example 2.8: $\frac{dE_m}{d\tau} = -\frac{3}{\tau^4}$. At $\tau = 0.062$: $-3/(0.062)^4 = -3/1.478 \times 10^{-5} \approx -2.03 \times 10^5$ (per second). At $\tau = 0.042$: $-3/(0.042)^4 \approx -9.64 \times 10^5$. A shortening of the PPTQ by 1 ms near the stress state raises E_m almost five times as strongly as the same shortening near the waking state — the law predicts a strongly nonlinear escalation of Mind-Energy with increasing arousal.

6.4 The combined model: PPTQ as bridge

From (5.11) it follows that $\tau = E_m^{-1/3}$ (power-rule inverse; see Rule 2). Substitution into (5.4) gives an RTC Kuramoto model in which Mind-Energy explicitly scales the dynamics:

$$\boxed{\frac{d\theta_i}{dP} = \frac{1}{E_m^{1/3}} \left[\omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i) \right]} \quad (5.13)$$

Herewith the PPTQ τ becomes the **formal bridge** between the Mind-Energy formulation and the resonance dynamics: one quantity (τ , or equivalently E_m) determines both the energy level of the mind and the rate at which percepts are formed.

Worked example 5.5. At $E_m = 4.20 \times 10^3$ (normal waking), $E_m^{1/3} = 16.13$, so $1/E_m^{1/3} = 0.062 = \tau$ — consistent. At $E_m = 1.35 \times 10^4$ (stress), $E_m^{1/3} = 23.8$, $1/E_m^{1/3} = 0.042 = \tau$. The prefactor in (5.13) is therefore literally the PPTQ, and a higher Mind-Energy compresses the phase change per percept.

6.5 The conceptual hierarchy

Resonance $\rightarrow$ PPTQ (τ) $\rightarrow$ perceptual succession (P) $\rightarrow$ phase dynamics (θ_i) $\rightarrow$ conscious experience.

The corresponding mathematical structure is the pair

$$E_m = \tau^{-3} \quad \text{and} \quad \frac{d\theta_i}{dP} = \tau(P) \left[\omega_i + \frac{K}{N} \sum_j \sin(\theta_j - \theta_i) \right].$$

7 PART VI — EPISTEMOLOGICAL ACCOUNTABILITY AND THE HARD PROBLEM

7.1 Four levels, sharply separated

RTC distinguishes four levels, and the strength of the formalism depends on not conflating them:

1. **Ontological level.** Experience is primary (Axiom I); temporal organisation is constitutive of experience. This is a metaphysical commitment; no experiment establishes it directly.
2. **Formal level.** τ becomes a formal variable; the relations $E_m \propto \tau^{-3}$ and $\frac{d\theta_i}{dP} \propto \tau$ impose structure.
3. **Operational level.** τ is defined as “the characteristic temporal interval over which successive perceptual states are integrated into a unitary experiential event”, and estimated psychophysically (temporal-order judgments, simultaneity judgments, temporal numerosity, backward masking, temporal-integration paradigms).
4. **Empirical level.** Only here does the theory become vulnerable to refutation.

The formalism *does not prove the axioms*. It converts them into a structured set of empirical consequences.

7.2 The status of the mathematical layers (Chapter 9)

Layer	Status	Function
Ginzburg–Landau field theory	heuristic	holographic field, phase transitions
Kuramoto synchronisation	formal + computational	percept formation, coherence
Bayesian inference	formal	perceptual updating
Nonlinear dynamics	heuristic + computational	attractor landscapes, bifurcations, psychopathology

Formal = deductive proof from accepted premises; *computational* = simulation validation; *heuristic* = structural analogy without literal physical identification. This modesty forestalls the classic error — mistaking analogy for proof — that has discredited many a formalisation in consciousness studies.

7.3 Why percept differentiation addresses the hard problem

The argument, explicit and step by step:

1. **The explanatory gap, in its usual formulation, is bound to clock time.** Standard theories describe neural processes as $\frac{d(\cdot)}{dt}$ — evolving in homogeneous external time. In that framework there is no place for the *lived* temporal structure; it is averaged away. The “what it is like to experience” then has no formal carrier.
2. **RTC introduces a second, psychophysically defined temporal variable.** Through $\frac{d(\cdot)}{dP}$ with $dt = \tau(P) dP$, the temporal organisation of experience acquires its own explicit formal carrier — $\tau(P)$ — that is *empirically distinguishable* from t .
3. **The shift makes the gap testable instead of mysterious.** The question “how does experience relate to physics?” is reformulated as: “Does a P -indexed (PPTQ-scaled) description explain the dynamics of conscious states *better* than a purely t -indexed description, after correcting for arousal, sensory input and overall neural power?” Formally: compare model M_P (equation 5.4) with model M_t (equation 4.3) in out-of-sample prediction. RTC predicts $M_P \succ M_t$.
4. **If $M_P \succ M_t$ holds up robustly, then the temporal organisation of experience is an autonomous explanatory variable** — not reducible to clock physics. That is the “mathematical

burden of proof” the thesis carries: not a proof that qualia exist, but an explicit, falsifiable formalism in which the distinction between physical time and experienced time does the explanatory work.

5. **What it does not claim.** It does not demonstrate that consciousness *is identical to* PPTQ, nor that resonance is ontologically fundamental. It opens a *route* by which those ontological claims acquire empirical consequences.

7.4 Falsifiability: the concrete criteria

- **Mind-Energy law.** $\ln E_m = \alpha - 3 \ln \tau + \epsilon$. Test $H_0 : \beta \neq -3$ against $H_1 : \beta = -3$. A Bayesian model comparison of $M_1 : \beta = -3$ against an unconstrained power-law model $M_2 : E_m = a\tau^\beta$ prevents an arbitrary power law from counting as “support”.
- **PPTQ-Kuramoto hypothesis.** If clock-indexed dynamics consistently give better predictions of perceptual and conscious-state transitions than PPTQ-indexed dynamics, the RTC modification loses its justification.
- **Avoiding circularity.** τ must be estimated psychophysically, while Mind-Energy and neural synchronisation are measured independently (EEG/MEG). Otherwise the test is not independent.

7.5 The epistemological transition summarised

metaphysical axiom $\rightarrow$ mathematical formalisation $\rightarrow$ operational definition $\rightarrow$ empirical prediction $\rightarrow$ falsifiable test.

Without this chain, RTC remains an ontological interpretation of consciousness. With it, RTC becomes a candidate scientific theory whose central claims can be confronted with data, compared with competing models, and rejected if its quantitative predictions fail.

8 PART VII — REFERENCE

8.1 Appendix A — All differentiation rules in one table

#	Rule	Formula
1	Constant	$(c)' = 0$
2	Power	$(x^n)' = nx^{n-1}$
3	Constant multiple	$(cf)' = cf'$
4	Sum / difference	$(f \pm g)' = f' \pm g'$
5	Product	$(fg)' = f'g + fg'$
6	Quotient	$(f/g)' = \frac{f'g - fg'}{g^2}$
7	Chain	$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$
8a	Sine	$(\sin x)' = \cos x$
8b	Cosine	$(\cos x)' = -\sin x$
8c	Tangent	$(\tan x)' = \sec^2 x$
8d	Exponential	$(e^x)' = e^x; (a^x)' = a^x \ln a$
8e	Logarithm	$(\ln x)' = 1/x$
8f	Inverse trig	$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}};$ $(\arctan x)' = \frac{1}{1+x^2}$
9	Implicit	differentiate $F(x, y) = 0$, solve for y'
10	Higher order	$f'' = (f')'$, etc.
11	Partial	$\partial f / \partial x_k$: other variables constant
12	Total (several vars.)	$\frac{df}{dt} = \sum_k \frac{\partial f}{\partial x_k} \frac{dx_k}{dt}$

8.2 Appendix B — Standard integrals

$f(x)$	$\int f dx$	$f(x)$	$\int f dx$
x^n ($n \neq -1$)	$\frac{x^{n+1}}{n+1} + C$	$\sin(kx)$	$-\frac{1}{k} \cos(kx) + C$
$1/x$	$\ln x + C$	$\cos(kx)$	$\frac{1}{k} \sin(kx) + C$
e^{kx}	$\frac{1}{k} e^{kx} + C$	$\frac{1}{1+x^2}$	$\arctan x + C$
$\ln x$	$x \ln x - x + C$	$\frac{1}{\sqrt{1-x^2}}$	$\arcsin x + C$

Techniques: substitution ($\int f(g)g' dx = \int f(u) du$); integration by parts ($\int u dv = uv - \int v du$); numerical (trapezoidal, Simpson, RK4).

8.3 Appendix C — List of symbols

Symbol	Meaning	Unit
t	homogeneous clock time	s
$\tau, \tau(P)$	PPTQ / experiential time quantum	s

Symbol	Meaning	Unit
P	perceptual succession / percept index	— (dimensionless)
θ_i	phase of oscillator i	rad
ω_i	intrinsic angular frequency of oscillator i	rad/s
K	coupling strength	rad/s
K_c	critical coupling strength	rad/s
N	number of oscillators	—
$R, R(P)$	Kuramoto order parameter (coherence)	$\in [0, 1]$
Ψ	collective / mean phase	rad
R_c	percept threshold (≈ 0.6)	—
E_m, M_E	Mind-Energy	normalised unit
v_τ	temporal velocity = $1/\tau$	s^{-1}
m_τ	effective psychophysical mass $\propto 1/\tau$	—
r	control parameter of the logistic map (reflexive amplification)	—
β	log-log regression slope (E_m vs. τ); RTC prediction -3	—
$\psi(x, t)$	Ginzburg-Landau order-parameter field	—
J	Jacobian matrix	—

8.4 Appendix D — Code examples

8.4.1 D.1 — Kuramoto (classical), R

```

kuramoto <- function(N = 50, K = 2.0, dt = 0.01, steps = 2000, sd_omega = 0.5) {
  omega <- rnorm(N, 0, sd_omega)
  theta <- runif(N, 0, 2 * pi)
  R_hist <- numeric(steps)
  for (s in seq_len(steps)) {
    z <- mean(exp(1i * theta))      # complex order parameter
    R <- Mod(z); Psi <- Arg(z)
    dtheta <- omega + K * R * sin(Psi - theta)  # mean-field form
    theta <- theta + dt * dtheta
    R_hist[s] <- R
  }
  R_hist
}
plot(kuramoto(K = 2.0), type = "l", ylim = c(0, 1),
     xlab = "iteration", ylab = "order parameter R")

```

8.4.2 D.2 — RTC Kuramoto (percept-indexed) with dynamic PPTQ, Python

```

import numpy as np

def tau_of_P(P):
    # PPTQ recovers from 43 ms (stress) toward 62 ms (waking)
    return 0.062 * (1 - 0.3 * np.exp(-P / 5.0))

def rtc_kuramoto(N=50, K=2.0, dP=0.05, steps=4000, sd_omega=0.5, seed=0):

```

```

rng = np.random.default_rng(seed)
omega = rng.normal(0.0, sd_omega, N)
theta = rng.uniform(0, 2*np.pi, N)
R_hist, P = [], 0.0
for _ in range(steps):
    z = np.mean(np.exp(1j * theta))
    R, Psi = np.abs(z), np.angle(z)
    tau = tau_of_P(P)
    dtheta = tau * (omega + K * R * np.sin(Psi - theta)) # equation (5.5)
    theta += dP * dtheta
    P += dP
    R_hist.append(R)
return np.array(R_hist)

```

8.4.3 D.3 — Mind-Energy law and log-log check, R

```

tau <- c(0.100, 0.080, 0.062, 0.050, 0.042) # PPTQ per state (s)
Em <- 1 / tau^3 # equation (5.11)
fit <- lm(log(Em) ~ log(tau))
coef(fit)[["log(tau)"]] # expected: -3

```

8.4.4 D.4 — Logistic map, iteration, R

```

logistic_iter <- function(r, x0 = 0.5, n = 40) {
  x <- numeric(n + 1); x[1] <- x0
  for (i in seq_len(n)) x[i + 1] <- r * x[i] * (1 - x[i])
  x
}
tail(logistic_iter(r = 3.2), 6) # settles into a 2-cycle ~0.513 / ~0.799

```

8.5 Appendix E — Strogatz reading guide by topic

Topic in this manual	<i>Infinite Powers</i>	<i>Nonlinear Dynamics and Chaos</i>	<i>Sync</i>
Limit, 0/0, derivative	ch. 4–6	§2.1–2.2	—
Integral, Fundamental Theorem	ch. 8–10	Appendix	—
Differential equations	ch. 10–11	ch. 2	—
Fixed points, linear stability	ch. 6	§2.4, ch. 5–6	—
Bifurcations (saddle-node, pitchfork, Hopf)	—	ch. 3, 8	ch. 5
Logistic map, period doubling, chaos	ch. 12	ch. 10	ch. 5
Phase, oscillators, phase response	—	ch. 4, 7	ch. 1–3
Two coupled oscillators, Adler equation	—	§4.5, §8.2	ch. 2
Kuramoto model, order parameter, K_c	—	§8.6	ch. 1–3
Synchronisation as a universal phenomenon	—	ch. 8	whole book

8.6 Appendix F — Cross-references to the thesis

Quantity / equation	Location in <i>The Resonating Self</i>
Axiom II (PPTQ)	Chapter 4
$v_\tau = dP/dt$; $M_E = \frac{1}{2}mv_\tau^2$	§4.9.1, §4.11 (revised equations 4.3–4.5)
Derivation $E_m \propto \tau^{-3}$; $E_m = 1/\tau^3$	§4.11 (revised equations 4.6–4.7); Chapter 9 §9.4 (equations 9.7–9.12)
Ginzburg–Landau functional and evolution	Chapter 9 §9.2 (equations 9.1–9.2)
Kuramoto model; order parameter	Chapter 9 §9.3 (equations 9.3–9.4)
τ -reparametrised Kuramoto ($dt = \tau(P) dP$)	Chapter 9 §9.4 (new in the revision of 30 Aug 2026)
Bayesian resonance engine; Fokker–Planck	Chapter 9 §9.5 (equation 9.13)
Logistic map; three phases; bifurcations	Chapter 9 §9.6 (equation 9.14); Figures 15–16
Core theorems (Synchrony, Criticality, Resonant Identity)	Chapter 9 §9.7
Falsifiability, $\beta = -3$, model comparison M_P vs. M_t	Chapter 14 (Empirical Testability)

9 Closing note

This manual has sought to demonstrate one thing: that the mathematical core of *The Resonating Self* — the differentiation of conscious dynamics with respect to the **percept** rather than with respect to **clock time** — can be built up entirely from the standard tools of calculus, with no mathematical tricks and no hidden assumptions. The chain rule (Rule 7) supplies the transition $\frac{d\theta}{dt} \rightarrow \frac{d\theta}{dP}$; the relation $dt = \tau(P) dP$ is an explicit, testable modelling assumption; the power rule (Rule 2) supplies the Mind-Energy law $E_m = \tau^{-3}$ with its falsifiable exponent $\beta = -3$.

What remains is not a proof that the *hard problem* has been solved in the sense that the mystery has vanished. What remains is a formalism in which the distinction between physical and experienced time performs the explanatory work, with predictions that can fail. In the spirit of Strogatz: calculus does not reveal *why* there is experience, but it does give us the language to say precisely *what* changes, with respect to *what*, and how fast — and that is the precondition for any science of consciousness worthy of the name.